

A Comprehensive Survey of Ensemble Clustering: Taxonomies, Methodologies, Experimental Benchmark, and Challenges

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Abstract—Ensemble clustering combines multiple base partitions into a consensus result to make clustering more robust and reliable. Over the past two decades, many methods have been proposed from different perspectives. However, the connections among these methods are still not well organized. Although several surveys have reviewed parts of the literature, most of them focus on particular method groups or application areas. A general view of problem settings, method relationships, empirical behavior, and open challenges is still lacking.

This paper presents a comprehensive survey of ensemble clustering from five perspectives: problem formulation, methodological taxonomy, application domains, experimental benchmarking, and future challenges. We first revisit the standard formulation and clarify several underexplored settings, including the relation to multi-view clustering, the use of original feature information, and incomplete-data scenarios. We then organize existing studies into three major categories, namely consensus modeling methods, ensemble reliability methods, and representation and scalability methods, which are further divided into seven subcategories. For each category, we summarize representative techniques, assumptions, strengths, and limitations. We also review major applications in biomedical analysis, healthcare, spatiotemporal monitoring, vision and multimedia, and textual and network analytics. To support systematic empirical comparison, we conduct a large-scale benchmark of 76 method variants on 86 datasets with eight evaluation metrics under a unified protocol. Finally, we discuss five future directions, including cluster-number determination, diversity enhancement, stability and robustness, interpretability, and fairness. Project page: <https://github.com/xuz2019/Ensemble-Clustering-Review>.

Index Terms—Ensemble clustering, taxonomy, benchmark, challenges, review.

I. INTRODUCTION

Clustering is a fundamental task in unsupervised learning and plays a central role in pattern analysis, data mining, and knowledge discovery. Over the past decades, a large number of clustering algorithms have been developed under different

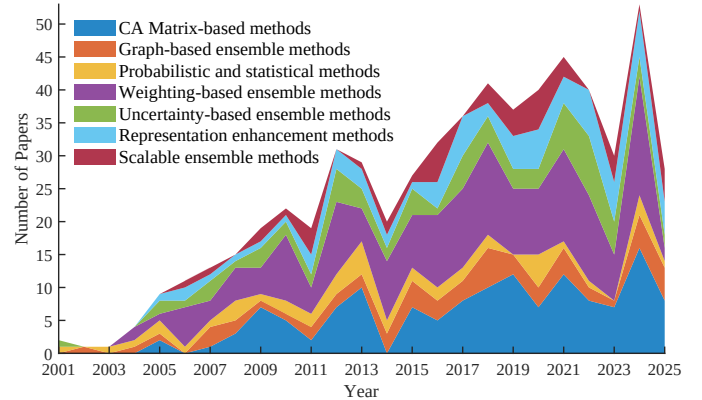


Fig. 1. Category-wise temporal trends of ensemble clustering research. The stacked area chart shows the annual number of papers in each of the seven methodological subcategories from 2001–2025. These subcategories jointly instantiate the three major categories in our unified taxonomy, and the total height represents the overall annual volume. Data were primarily collected from DBLP.

assumptions on data distributions and optimization objectives [1], [2], [3]. Despite their success, clustering results are often highly sensitive to the choice of algorithm, hyper-parameters, initialization, and data representation. Consequently, even on the same dataset, different clustering algorithms may produce substantially different partitions, raising concerns about the stability of a single clustering solution.

To address this issue, ensemble clustering [4], also known as consensus clustering, has emerged as an effective approach for integrating multiple base clusterings into a consensus partition. Different base clusterings may reveal complementary structures, and their aggregation can transform such diversity into a more robust clustering result. Empirical evidence also shows that combining multiple partitions often leads to better performance than relying on a single clustering algorithm [5], [6]. Accordingly, ensemble clustering has attracted sustained attention and has been applied in a variety of domains, including biomedical omics analysis [7], healthcare analytics [8], temporal and spatiotemporal monitoring [9], vision and multimedia analysis [10], textual and network analytics [11].

Compared with supervised ensemble learning, ensemble clustering is intrinsically more challenging. Ground-truth labels are unavailable, so the aggregation process cannot be directly guided by external supervision. In many cases, only the outputs of base clusterings are observed, with no direct

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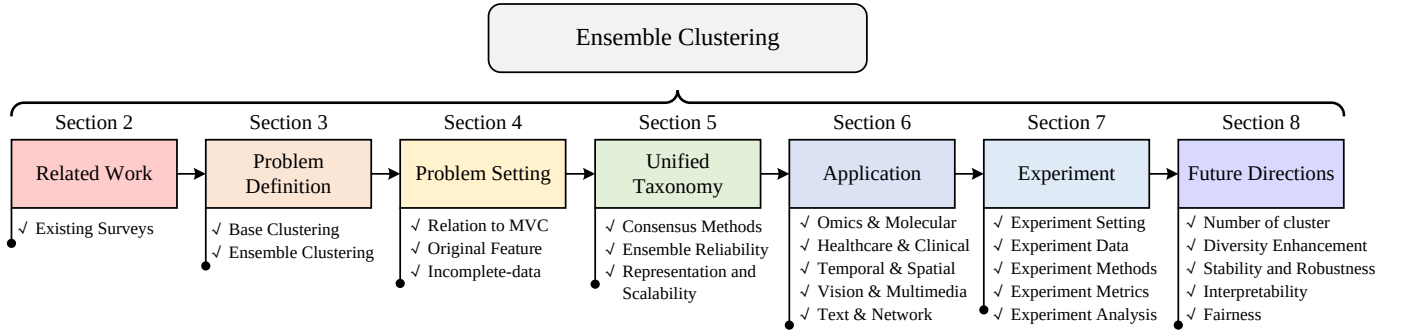


Fig. 2. The structure of this paper

access to the original feature space. These characteristics make principled consensus construction difficult. Existing studies therefore differ in the type of ensemble evidence they emphasize, including pairwise co-occurrence, graph or hypergraph structure, and probabilistic relations in the partition space. They also differ in the stage of the ensemble pipeline they aim to improve, ranging from consensus construction and evidence reliability estimation to scalable optimization and representation enhancement. Fig. 1 shows the publication trends of different types of methods since 2001.

Although several surveys on ensemble clustering have been published [12], [13], [14], [15], [16], their coverage remains fragmented. Existing reviews mainly focus on specific aspects, such as classical consensus mechanisms, ensemble selection, or weighting strategies, and differ substantially in their treatment of data settings, application scenarios, and review scope. In particular, the literature still lacks a comprehensive review that jointly synthesizes problem settings, methodological families, application scenarios, and empirical behavior under a unified perspective. More importantly, existing surveys rarely provide large-scale, reproducible benchmarking results, making it difficult for the community to compare different methods systematically and to identify reliable directions for future research.

In this paper, we present a comprehensive survey of ensemble clustering to provide a systematic overview of existing studies and support future research. We revisit the basic formulation of ensemble clustering and clarify several overlooked problem settings, including its relation to multi-view clustering, the use of original feature information, and incomplete-data scenarios. We then develop a unified taxonomy that organizes existing methods into three major categories, namely Consensus Modeling Methods, Ensemble Reliability Methods, and Representation and Scalability Methods, which are further divided into seven subcategories. For each category, we summarize representative techniques, assumptions, strengths, and limitations, and we further review representative application domains. To support systematic empirical comparison, we build a large-scale benchmark involving 76 method variants, 86 datasets, and eight evaluation metrics under a unified protocol. Finally, we discuss five future directions, including cluster-number determination, diversity enhancement, stability and robustness, interpretability, and fairness.

The main contributions of this survey are summarized as

follows.

- We provide a more comprehensive and up-to-date review of ensemble clustering by jointly considering problem formulation, methodology, applications, experiments, and future challenges, where problem formulation covers both the basic definition and extended problem settings.
- We organize existing studies into three major methodological categories and seven subcategories, and clarify the connections, assumptions, strengths, and limitations of different research lines under a unified framework.
- We present a large-scale empirical benchmark over 76 method variants and 86 datasets, offering a systematic comparison of different categories of ensemble clustering methods.
- We identify five open problems and promising directions for next-generation ensemble clustering methods, including the determination of the cluster number, diversity enhancement, stability and robustness, interpretability, and fairness.

The rest of this survey is organized as illustrated in Fig. 2. Section II reviews related work and existing surveys. Section III introduces the problem definition of ensemble clustering, including the fundamentals of base clustering and ensemble clustering. Section IV presents the problem settings from multiple perspectives. Section V then provides a unified taxonomy of ensemble clustering methods. Following this methodological discussion, Section VI summarizes major application domains, and Section VII reports experimental settings, results, parameters, and analyses on representative methods. Section VIII discusses future challenges of ensemble clustering, and the paper is concluded in the final section.

II. RELATED WORK

Ensemble clustering has been reviewed from several complementary perspectives over the past decade. Early surveys mainly established the classical formulation of ensemble generation and consensus design, whereas later studies tended to focus on more specialized topics, such as ensemble selection or weighting.

Vega-Pons and Ruiz-Shulcloper [12] provided one of the earliest comprehensive reviews of clustering ensemble algorithms. Their survey emphasized the classical consensus-modeling perspective and covered representative families such

TABLE I
COMPARISON OF EXISTING SURVEYS AND OUR REVIEW FROM DATA-SETTING, METHODOLOGICAL, APPLICATION, AND SCOPE PERSPECTIVES

Category	Item	Vega-Pons [12]	Boongoen [13]	Golalipour [14]	Zhang [15]	Wang [16]	Ours
Problem Setting	Relation to multi-view clustering	–	–	–	–	–	✓
	Discusses original feature information ¹	–	–	–	–	–	✓
	Discusses incomplete-data settings	–	○	○	○	○	✓
Methodology	<i>Consensus Modeling Methods</i>						
	CA matrix-based methods	✓	✓	○	✓	–	✓
	Graph-based ensemble methods	✓	✓	○	○	✓	✓
	Probabilistic and statistical methods	✓	○	–	–	✓	✓
	<i>Ensemble Reliability Methods</i>						
	Weighting-based ensemble methods	–	○	✓	✓	–	✓
	Uncertainty-based ensemble methods	○	–	○	✓	○	✓
	<i>Representation and Scalability Methods</i>						
	Representation enhancement methods	–	–	–	○	○	✓
Application	Scalable ensemble methods	○	✓	○	○	○	✓
	Biomedical Omics Analysis	✓	✓	–	○	✓	✓
	Healthcare Analytics and Decision Support	○	✓	–	–	✓	✓
	Temporal and Spatiotemporal Monitoring	–	✓	✓	✓	–	✓
	Visual and Multimedia Data Analysis	✓	○	✓	–	○	✓
Review Scope	Textual, Network, and Behavioral Analytics	○	○	○	○	✓	✓
	Provides comparative experiments	–	–	–	–	○	✓
Review Scope	Discusses future directions	✓	✓	✓	✓	○	✓

Note: ✓ indicates systematic discussion; ○ indicates partial coverage or brief mention; – indicates not explicitly discussed.

as co-association-based, graph-based, and probabilistic formulations. This work laid an important conceptual foundation for subsequent studies, but it mainly reflects the early development stage of the field.

Boongoen and Iam-On [13] expanded the scope by reviewing ensemble generation strategies, ensemble representations, consensus functions, applications, and research challenges. While their survey extends significantly into practical domains and scalability considerations, its organization still largely follows the classical taxonomy centered on consensus mechanisms, with limited discussion of recent directions such as uncertainty-aware and representation-enhancement methods.

Golalipour et al. [14] focused on clustering ensemble selection and highlighted the role of quality and diversity in selecting informative base clusterings. Their review is particularly useful for understanding selection-oriented optimization, but its scope is intentionally narrower than that of a general survey on clustering ensemble methodologies.

Zhang [15] further specialized the discussion to weighted clustering ensemble. This review systematically summarized weighting strategies at different levels, including clusterings, clusters, features, and objects, and discussed applications to multi-view and temporal data. While highly informative for weighting-based models, it does not attempt to unify weighting with other major methodological families under a broader survey framework.

Related but distinct from clustering ensemble, Wang et al. [16] recently surveyed co-clustering methods from the perspectives of information theory, graph theory, matrix factorization, and other related paradigms. They further summarized representative applications and provided comparative experiments on benchmark datasets. Although co-clustering differs from clustering ensemble in formulation and objective, this survey offers useful complementary insights into taxonomy

construction, application-oriented analysis.

The differences among these studies are also summarized in Table I. In summary, existing reviews provide valuable yet partial perspectives on the field, with most focusing on only one part of the landscape, such as classical consensus functions, ensemble selection, weighting strategies, or related clustering paradigms. In addition, few prior reviews offer a systematic comparative benchmark, in-depth experimental analysis, or reproducible resources such as unified code and evaluation protocols for cross-method comparison. As a result, a unified review that jointly covers data settings, methodological families, application scenarios, review scope, and large-scale empirical comparison remains lacking.

III. PROBLEM DEFINITION

Let $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$ denote a dataset of n objects, where, in the standard setting, each object x_i is represented in a feature space $\mathcal{F} \subseteq \mathbb{R}^d$.

A. Base Clustering

A base clustering (or base partition) is a partition of $\mathcal{X}$ produced by a clustering algorithm. The m -th base clustering can be written as

$$\pi^{(m)} = \{C_1^{(m)}, C_2^{(m)}, \dots, C_{k_m}^{(m)}\},$$

¹Traditional ensemble clustering typically derives a consensus partition solely from the outputs of base clusterings, without directly accessing the original feature space. However, an increasing number of recent studies have incorporated original feature information into the ensemble process to improve representation quality, while this distinction has rarely been explicitly discussed in existing survey papers.

TABLE II
COMPARISON BETWEEN MULTI-VIEW CLUSTERING AND ENSEMBLE CLUSTERING

Dimension	Multi-view Clustering	Ensemble Clustering
Common idea	Improve clustering robustness and accuracy	Improve clustering robustness and accuracy
Input	Multiple raw features or representations	Multiple clustering partitions
Feature access	Usually required	Usually not required
Fusion level	Early, middle and late fusion	Usually related to late fusion
Modeling granularity	Usually relies on a global optimization framework	Often focuses on finer partition-level, cluster-level structure
Typical use case	Multi-modal or multi-representation data	Robust aggregation of multiple clusterings

where k_m is the number of clusters and each $C_j^{(m)} \subseteq \mathcal{X}$ denotes a cluster. For hard clustering, the clusters satisfy

$$\bigcup_{j=1}^{k_m} C_j^{(m)} = \mathcal{X}, \quad C_j^{(m)} \cap C_{j'}^{(m)} = \emptyset, \quad j \neq j'.$$

Equivalently, $\pi^{(m)}$ may be represented as a cluster assignment function which maps each data point to a cluster index

$$\pi^{(m)} : \mathcal{X} \rightarrow \{1, 2, \dots, k_m\},$$

or as an n -dimensional label vector

$$\pi^{(m)} = (y_1^{(m)}, y_2^{(m)}, \dots, y_n^{(m)})^\top,$$

where $y_i^{(m)} \in \{1, 2, \dots, k_m\}$ denotes the cluster assignment (or cluster index) of sample x_i . Note that these labels are nominal and do not carry intrinsic ordering.

The collection of M base clusterings can be represented as

$$\Pi = \{\pi^{(1)}, \pi^{(2)}, \dots, \pi^{(M)}\}.$$

In practice, these base clusterings may be generated by different clustering algorithms, parameter settings, initializations, feature subsets, sampled instances, or data views. Such variations introduce diversity into the clustering ensemble, which is generally regarded as an important source of robustness for the final consensus result.

B. Ensemble Clustering

The goal of ensemble clustering is to infer a consensus partition from Π . Formally, one seeks a clustering

$$\pi^* = \{C_1^*, C_2^*, \dots, C_K^*\}, \quad \pi^* \in \mathcal{P}(\mathcal{X}),$$

where K denotes the number of clusters in the consensus partition, $\mathcal{P}(\mathcal{X})$ denotes the set of all possible partitions of $\mathcal{X}$, such that π^* best summarizes the information contained in the ensemble.

From a general viewpoint, ensemble clustering can be expressed as

$$\pi^* = f(\pi^{(1)}, \pi^{(2)}, \dots, \pi^{(M)}),$$

where $f(\cdot)$ is a consensus function that aggregates base clusterings and outputs a final partition.

Note that cluster labels are permutation-invariant: the label “1” in one base clustering does not necessarily correspond to the label “1” in another. Moreover, base clusterings may differ

in the number of clusters, the underlying clustering principles, and their quality, diversity, and redundancy. Therefore, the design of $f(\cdot)$ is nontrivial and lies at the core of ensemble clustering.

IV. PROBLEM SETTINGS

Existing surveys have primarily focused on methodological taxonomies, while paying less attention to the problem settings of ensemble clustering. In this section, we examine several setting-related issues that remain insufficiently clarified in the literature, including the relation of ensemble clustering to multi-view clustering (MVC), the emergence of feature-enhanced ensemble clustering, and incomplete-data scenarios. Our goal is to provide a clearer conceptual view of ensemble clustering from the perspective of data settings.

A. Relation to Multi-view Clustering

Although MVC and ensemble clustering share substantial conceptual overlap, they are not fully equivalent. MVC typically takes multiple raw features or data representations as input, whereas ensemble clustering usually operates on multiple clustering partitions and therefore does not necessarily require access to the original feature space. From the perspective of information fusion, MVC may involve early, middle, or late fusion (for a more comprehensive discussion of MVC, readers are referred to the recent survey [17]), while ensemble clustering is more closely related to the late-fusion setting. They also differ in methodological emphasis: MVC often focuses on jointly modeling multiple views within a global optimization framework, whereas ensemble clustering pays more attention to partition-level and cluster-level relationships among base clusterings. As a result, although MVC methods can sometimes be adapted to ensemble clustering, they often perform less effectively than algorithms specifically designed for the ensemble clustering setting. Their main similarities and differences are summarized in Table II.

B. Use of Original Features in Ensemble Clustering

In the initial literature [4] on ensemble clustering, the problem was commonly defined as combining multiple partitionings of a set of objects without accessing the original features. Under this setting, many influential studies were subsequently developed. For example, Fred [6] introduced the concept of evidence accumulation, which laid a solid

foundation for the co-association matrix-based line of research in ensemble clustering. Topchy [18] proposed a mixture-of-multinomials model and solved it using the EM algorithm, further showing that even weak clusterings can yield strong performance when combined through an ensemble.

However, an increasing number of studies have begun to incorporate original feature information into the ensemble process, rather than relying solely on base partitions. For instance, LiteWSEC [19] leverages the original feature space for representative sampling and affinity construction, while HSCE [20] further exploits local neighborhood information from the original data to improve consensus learning. This tendency is more evident in representation enhancement ensemble clustering [21], [22], [23], where original features are commonly used to learn latent representations. Such settings are particularly valuable when base partitions alone are insufficient to capture the intrinsic structure of the data.

At the same time, the use of original features does not necessarily make these methods instances of MVC. We instead regard them as feature-enhanced ensemble clustering, because their primary objective remains the aggregation of multiple base partitions into a consensus partition.

C. Incomplete-Data Settings

Incomplete-data studies related to ensemble clustering can be divided into two settings. The first is inherited from incomplete multi-view clustering, where missingness occurs in the input space, such as missing views, incomplete kernels, or partial data representations. Methods such as CRTCC [24], EE-IMVC [25], and MIC [26] address this problem by recovering or completing these inputs before or during consensus learning. Although closely related, this setting does not characterize missingness in ensemble clustering itself.

The second setting is intrinsic to ensemble clustering: the original features may be unavailable, and incompleteness occurs directly in the base partitions, so that some objects are unlabeled in certain clusterings or different clusterings cover different subsets of samples. In this case, the missing information lies in the consensus evidence rather than in the raw data, which makes co-association estimation, cluster correspondence, and evidence weighting substantially more difficult. Zhou et al. [27] provided an early representative study by jointly imputing incomplete base partition matrices and learning consensus without using the original feature space. We view this partition-level incomplete setting as an emerging direction that deserves dedicated study, rather than as a simple extension of incomplete MVC.

V. A UNIFIED TAXONOMY OF ENSEMBLE CLUSTERING

Existing ensemble clustering methods differ both in their technical mechanisms and the stage of the ensemble pipeline. According to this criterion, we organize them into three major categories, as illustrated in Fig. 3.

Due to space limitations, this section provides a high-level methodological analysis of each category in our taxonomy. For each line of research, we focus on its main technical rationale, representative strengths, and key limitations. A more

detailed discussion of representative methods is provided in the supplementary material, which readers may consult for a more comprehensive understanding.

A. Consensus Modeling Methods

Consensus modeling methods form the classical core of ensemble clustering. Their goal is to infer a stable consensus partition from multiple base partitions that may be noisy, redundant, and label-permuted. Since the original feature space is often unavailable in the standard setting, these methods mainly rely on the structural evidence contained in the ensemble itself. The following discussion is organized into three lines according to the form of evidence used for consensus construction: pairwise agreement, graph structure, and probabilistic latent variables.

1) *CA Matrix-based Methods*: CA matrix-based methods avoid the need to align cluster labels across different base partitions. They transform each base clustering into pairwise co-occurrence evidence by recording whether two samples appear in the same cluster. Aggregating such evidence over the ensemble produces a consensus similarity matrix, where a larger value indicates stronger support for assigning the two samples to the same consensus structure.

Formally, following [6], the CA matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is defined as

$$\delta_{ij}^{(m)} = \begin{cases} 1, & \text{if } \pi^{(m)}(x_i) = \pi^{(m)}(x_j) \\ 0, & \text{otherwise,} \end{cases} \quad \mathbf{A}_{ij} = \frac{1}{M} \sum_{m=1}^M \delta_{ij}^{(m)},$$

where $\mathbf{A}_{ij} \in [0, 1]$ is the proportion of base clusterings in which x_i and x_j are assigned to the same cluster. Once the CA matrix is obtained, an off-the-shelf clustering method such as hierarchical clustering or spectral clustering is applied to derive the final consensus partition.

This formulation assumes that pairwise co-occurrence can serve as an informative surrogate for the latent consensus structure. By working directly at the sample-pair level, CA-based methods avoid the need to establish label correspondence across different base partitions. They also provide a flexible interface for heterogeneous ensembles, since any base clustering that outputs a partition can contribute to $\mathbf{A}$. These properties explain why CA-based methods remain one of the strongest classical paradigms in practice.

At the same time, the representation is deliberately coarse. Once the ensemble is compressed into $\mathbf{A}$, higher-order relations among clusters, neighborhoods, and entire base partitions are partially lost. Different ensembles may induce similar pairwise averages while implying different global structures. In addition, dense CA matrices require $O(n^2)$ storage and are sensitive to noise accumulation when many ensemble members are redundant or unreliable. A second-stage clustering algorithm is also still needed, so the final result depends both on the quality of the CA matrix and on the behavior of the downstream partitioner.

Work in this line has gradually moved from direct accumulation toward more structured CA estimation. Early studies mainly accumulated pairwise votes and reclustered the resulting matrix [6]. Later refinements exploited inter-cluster links

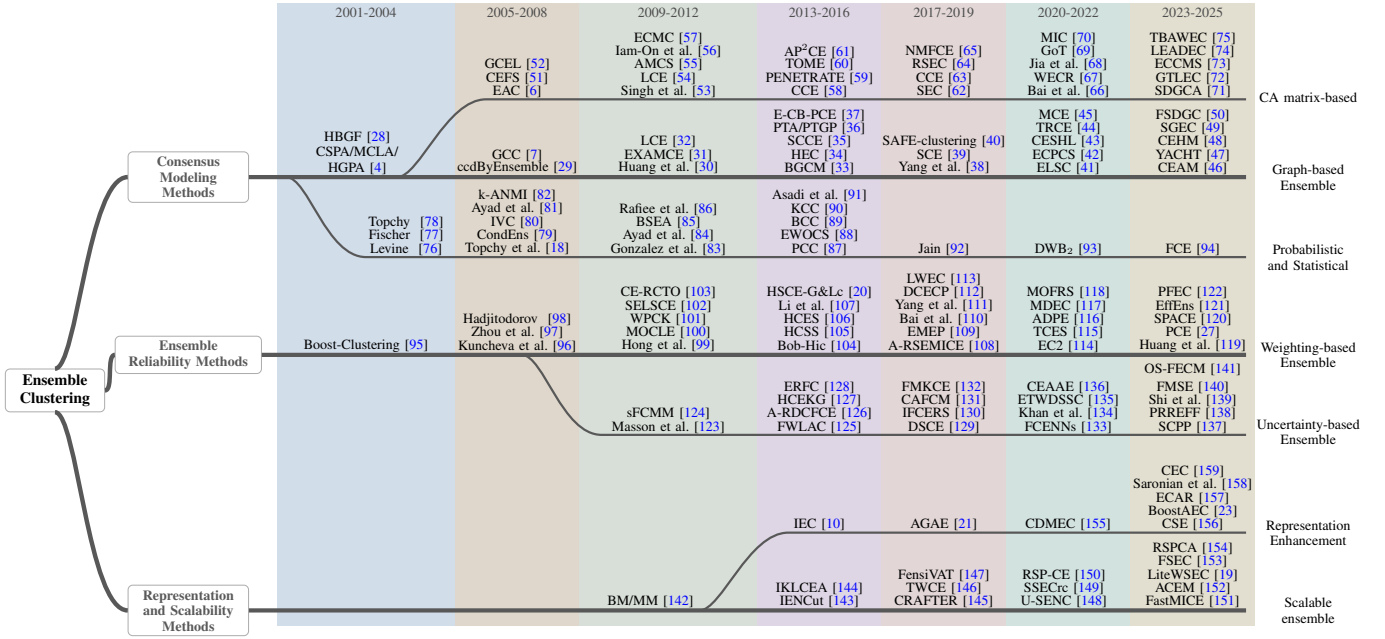


Fig. 3. A unified taxonomy of ensemble clustering methods (2001–2025). Existing ensemble clustering methods are organized into three major categories and seven subcategories, highlighting their development over time.

[54], path-based propagation [56], or density-aware structural correction [160], making pairwise evidence less local and less sparse [60], [69]. Another group of methods recast CA construction as recovery or optimization, using low-rank modeling [161], matrix completion [57], semidefinite optimization [53], weighted spectral reformulation [62], or semi-supervised regularization [67] to suppress unreliable evidence and recover a more faithful consensus similarity.

More recent methods extend the pairwise view through coupling [58], tensor structure [72], high-order topology [162], latent-space regularization [74], similarity-dissimilarity co-modeling [71], and balance constraints [75]. These developments do not replace the original CA intuition; they expose how much the field now expects from the pairwise representation. A useful CA matrix should be label-invariant, robust to noisy or redundant base partitions, expressive enough to preserve higher-order structure, and efficient enough for modern data scales. This maturity also raises the bar for future contributions. Generic refinements of the CA matrix are most compelling when they address a clearly identified failure mode, improve scalability, or provide stronger theoretical guarantees. Promising directions include scalable CA modeling, efficient high-order structure preservation, principled uncertainty or reliability estimation for pairwise evidence, and theory explaining when a refined consensus matrix is preferable to the raw one.

2) *Graph-based Ensemble Methods*: Graph-based ensemble methods start from a broader view of ensemble evidence. Consensus is not only a matter of whether two samples co-occur; it also depends on how local neighborhoods, clusters, partitions, and their higher-order relations are organized. These methods therefore translate ensemble clustering into graph construction followed by graph partitioning: the ensemble is

used to infer a consensus graph, and the final partition is recovered by cutting or factorizing that graph rather than by directly reclustering a CA matrix.

Given a base clustering ensemble Π , these methods typically learn a consensus graph by solving

$$\mathbf{W}^* = \arg \min_{\mathbf{W} \in \mathcal{S}} \mathcal{L}(\mathbf{W}; \Pi) + \lambda \mathcal{R}(\mathbf{W}),$$

where $\mathbf{W} \in \mathbb{R}^{n \times n}$ is the affinity matrix of the consensus graph for n data instances, $\mathcal{S}$ denotes the feasible set of admissible graphs, $\mathcal{L}(\mathbf{W}; \Pi)$ measures the discrepancy between $\mathbf{W}$ and the ensemble information encoded in Π , $\mathcal{R}(\mathbf{W})$ is a regularization term, and $\lambda > 0$ controls the trade-off between data fidelity and regularization. The final consensus clustering is then obtained by partitioning the learned graph $\mathbf{W}^*$ using an appropriate graph partitioning criterion.

This perspective relies on the assumption that the latent consensus structure can be faithfully encoded by a graph, hypergraph, or bipartite graph, and that graph partitioning objectives are aligned with the desired clustering. Under this assumption, graph-based methods solve several issues that are harder for plain CA aggregation: they can emphasize connectivity, neighborhood reliability, and higher-order relations; they can jointly model samples and clusters; and they naturally support weighted, multiplex, or hypergraph fusion.

The trade-off is that graph quality becomes the new bottleneck. These methods are sensitive to how edges, neighborhoods, anchors, or hyperedges are defined, and many formulations still depend on expensive spectral decomposition or iterative graph learning. In practice, their extra expressive power often comes with more design choices, more hyperparameters, and a more difficult scalability problem than in classical pairwise consensus schemes.

The development of graph-based ensemble clustering closely reflects these trade-offs. Early methods mainly treated consensus as direct graph partitioning, including similarity-graph, hypergraph, and bipartite formulations such as CSPA, HGPA, MCLA [4], and the hybrid bipartite graph model [28]. Later work shifted attention from cutting a graph to constructing a reliable one, leading to random-walk similarity [36], elite-neighbor refinement [42], and other structural denoising strategies. More recent methods learn the graph adaptively from the ensemble itself, often through unified graph optimization [163], [44], multiplex diffusion [46], or multi-graph fusion [164].

In parallel, tensorized [165], hypergraph [43], and anchor-based variants [47] have been introduced to encode higher-order structure while reducing the cost of full dense graph learning [48]. Viewed in this way, graph-based methods form a structural generalization of CA-based thinking: they preserve pairwise evidence but embed it in a richer connectivity model. Their practical strength suggests that graph structure, cluster-level propagation, and learned connectivity often capture information that plain pairwise averaging misses. The theoretical picture, however, is still incomplete. Performance depends heavily on graph quality, yet the field lacks a general account of why a particular construction, propagation scheme, or learned structure should remain reliable under noisy base partitions. Additional graph layers, regularizers, propagation steps, or weight terms are therefore most valuable when they clarify a specific ensemble failure mode rather than merely increasing architectural complexity. Future progress is likely to rely on scalable learned graph construction, reliability-aware graph learning, and robust graph quality estimation.

3) *Probabilistic and Statistical Methods*: Probabilistic and statistical methods give consensus clustering an explicit inferential meaning. Instead of treating base partitions merely as votes to be counted, they model them as noisy observations generated from latent consensus variables. This view is useful when one wants not only a final partition, but also an explanation of why that partition is plausible and how disagreement among base clusterings should be interpreted.

A classical formulation in this line was proposed by Topchy et al. [78]. Given an ensemble of M base clusterings, each sample x_i is represented by its label vector

$$\mathbf{y}_i = (\pi^{(1)}(x_i), \pi^{(2)}(x_i), \dots, \pi^{(M)}(x_i)),$$

so that consensus clustering is transformed into clustering the discrete observations $\{\mathbf{y}_i\}_{i=1}^n$. They model $\mathbf{y}_i$ by a finite mixture of multinomial distributions,

$$P(\mathbf{y}_i | \Theta) = \sum_{c=1}^K \alpha_c P_c(\mathbf{y}_i | \theta_c),$$

where K is the number of consensus clusters, α_c is the mixing weight of component c , and P_c is the corresponding component distribution. Under a conditional independence assumption across base clusterings,

$$P_c(\mathbf{y}_i | \theta_c) = \prod_{j=1}^M P_c^{(j)}(y_{ij} | \theta_c^{(j)}),$$

with each $P_c^{(j)}$ modeled as a multinomial distribution over the labels of the j -th base clustering. The parameters are estimated by maximizing

$$\log L(\Theta | Y) = \sum_{i=1}^n \log \sum_{c=1}^K \alpha_c P_c(\mathbf{y}_i | \theta_c)$$

via the EM algorithm, and the final consensus label is obtained from the component with the largest posterior probability.

The strength of this paradigm is that it makes the label space itself the modeling object. As a result, label correspondence across base clusterings is handled implicitly by latent variables rather than by explicit relabeling heuristics. This line also offers a principled way to quantify uncertainty, attach likelihoods or posterior probabilities to consensus assignments, and sometimes reason about the number of clusters through model evidence or priors.

These advantages, however, depend on stronger assumptions than in most classical consensus functions. Mixture-based models often assume conditional independence or specific noise forms across base clusterings, which may be unrealistic when the ensemble contains highly correlated partitions. Parametric misspecification, local optima in EM-style fitting, sensitivity to priors, and additional computational overhead are therefore recurring limitations of the category. In short, probabilistic methods gain interpretability and uncertainty quantification by paying a price in modeling assumptions and optimization complexity.

Subsequent work broadened the statistical view in several directions. One branch extended the original frequentist perspective through EM-like median partition search and utility-based formulations, showing that consensus learning can be cast as structured inference in partition space [80], [90]. Another made uncertainty modeling more explicit through Bayesian consensus clustering [89], probabilistic evidence-accumulation interpretations [87], and graphical-model formulations such as factor graphs [166]. A third line focused less on a particular generative model and more on the statistical foundations surrounding consensus, including resampling-based stability [77], theoretical consistency [92], information-theoretic interpretation [82], and systematic label-alignment analysis [81], [167].

The importance of probabilistic and statistical methods lies in a question that other paradigms often leave implicit: what probabilistic meaning, if any, should be attached to a consensus partition? This category is therefore best understood as foundational rather than purely accuracy-oriented. Its mature component is the basic probabilistic formulation of consensus; its less mature component is the theory of finite ensembles, uncertainty calibration, and robust inference under correlated or biased base clusterings. Further EM-style mixture models, Bayesian consensus variants, or label-alignment probabilistic updates are most useful when they bring new theoretical guarantees, calibrated uncertainty propagation, soft consensus that changes downstream decisions, fairness- or constraint-aware probabilistic fusion, or multi-source settings where probabilistic semantics are necessary rather than optional.

B. Ensemble Reliability Methods

Evidence reliability optimization methods ask a different question from consensus modeling methods. Instead of asking how to aggregate the ensemble, they ask whose evidence in the ensemble should count more. Their motivation is that a clustering ensemble is rarely homogeneous: some base partitions are reliable, others are redundant or unstable, and even within one partition different clusters or samples may have very different credibility. These methods therefore focus on evaluating, filtering, weighting, or uncertainty-modeling ensemble evidence before or during consensus learning.

Within this category, weighting-based and uncertainty-based methods are related but not identical. Weighting methods mainly ask *which* partitions, clusters, objects, or pairs should be trusted more. Uncertainty-based methods mainly ask *where* the consensus process should hesitate, soften, or defer a hard decision. This distinction is important because reliability or uncertainty is most meaningful when it changes how evidence is used, rather than only adding another score to an otherwise standard consensus pipeline.

1) *Weighting-based Ensemble Methods*: Weighting-based ensemble methods challenge the default assumption of uniform aggregation. In practice, base clusterings rarely contribute equally useful evidence: some are unstable, some are redundant, and some capture only local structure. Weighting methods try to estimate these differences at the level of partitions, clusters, objects, or sample pairs, using indirect signals such as internal agreement, diversity, stability, local consistency, or structural compatibility when ground-truth labels are unavailable.

A major development in this line is the transition from *global* weighting of whole base clusterings to *local* weighting of individual clusters or objects. A representative example is *Locally Weighted Ensemble Clustering* (LWEC) [113], which estimates cluster reliability according to how consistently the objects of a cluster are redistributed across the ensemble:

$$H(C_\ell) = \sum_{m=1}^M H^{(m)}(C_\ell), \quad H^{(m)}(C_\ell) = - \sum_{j=1}^{k_m} p_j \log_2 p_j,$$

where $H^{(m)}(C_\ell)$ denotes the uncertainty of cluster C_ℓ with respect to the base clustering $\pi^{(m)}$, measured by Shannon entropy, and $p_j = p(C_\ell, C_j^{(m)})$ denotes the proportion of objects in C_ℓ that fall into cluster $C_j^{(m)}$ in the m -th base clustering. Based on this uncertainty, LWEC defines the ensemble-driven cluster index

$$\text{ECI}(C_\ell) = \exp\left(-\frac{H(C_\ell)}{\theta M}\right),$$

where $\theta > 0$ controls the influence of cluster uncertainty on the index. The conventional co-association matrix is then refined into a locally weighted form,

$$\tilde{a}_{ij} = \frac{1}{M} \sum_{m=1}^M \text{ECI}\left(C^{(m)}(x_i)\right) \delta_{ij}^{(m)},$$

where $C^{(m)}(x_i)$ denotes the cluster containing x_i in the m -th base clustering. When $\delta_{ij}^{(m)} = 1$, x_i and x_j belong to the same

cluster, so the ECI term is a cluster-level reliability weight rather than a sample-specific weight. Co-occurrences supported by reliable clusters therefore contribute more strongly to the final consensus.

Weighting-based methods solve an important practical problem: they reduce the influence of low-quality or redundant evidence without forcing the field to abandon existing consensus mechanisms. This is why they are often easy to combine with CA-matrix or graph-based methods. However, their main weakness is equally clear. In unsupervised settings, quality surrogates are only proxies, and high agreement does not necessarily imply correctness. Weighting can therefore reward conformity rather than informativeness, suppress minority but meaningful structures, and create feedback loops in which dominant ensemble members become even more dominant. Additional scoring criteria and hyperparameters may also increase instability across datasets.

Research in this category has moved through several stages. Early work emphasized adaptive resampling and boosting-style reweighting, encouraging subsequent clusterers to focus on hard or unstable samples [168], [96]. Later studies tied weighting more closely to diversity generation: a weight is informative only when the ensemble contains genuinely complementary alternatives, which motivated instance-level [103] and attribute-level perturbation [169], generator design [170], localized alternative ensembles [171], and meta-clustering perspectives [172]. A further step introduced explicit selection and pruning [97], where weak members are removed rather than merely downweighted [106], [173]. Recent formulations often cast weighting and selection as multi-objective [100], [109] or transfer-aware optimization problems [115], so that reliability and diversity are considered jointly rather than heuristically [174].

Taken together, weighting methods are better viewed as reliability-control and robustness mechanisms than as universally dominant standalone consensus engines. The field has moved from the practical question of how to assign weights to the deeper question of how to estimate reliability without confusing agreement, diversity, and true usefulness. Since the core insight of selective trust is already mature, small variations of heuristic scalar scores based on entropy, diversity, NMI-like agreement, or quality-diversity sums are unlikely to be sufficient on their own. More substantive progress will depend on modeling reliability as a principled property of the ensemble process itself, rather than as an auxiliary score appended before aggregation.

2) *Uncertainty-based Ensemble Methods*: Uncertainty-based ensemble methods give disagreement a different interpretation. Inconsistency among base partitions is not always noise to be suppressed; it may also indicate intrinsic ambiguity in the data. This perspective is especially important when clusters overlap, boundary samples are abundant, or the data are noisy and heterogeneous, because forcing every object into an equally confident hard assignment can obscure useful uncertainty.

The key design choice is to represent uncertainty explicitly rather than collapse it too early into a hard label. In doing so, uncertainty-based methods preserve ambiguous evidence that

hard consensus methods often discard.

This benefit comes with clear trade-offs. Soft outputs are often more complex to optimize, interpret, and evaluate than crisp consensus labels. The final hard decision may still depend on thresholds or post-processing rules, and the field lacks the same degree of standardization in evaluation that classical hard-partition ensemble methods enjoy. Moreover, the additional uncertainty machinery can increase both the number of modeling choices and the computational burden.

The meaning of “uncertainty” has expanded over time. Early work focused on fuzzy consensus and soft-membership preservation [175], with recent one-step frameworks [141] and reliability-aware fuzzy ensembles [176] further integrating heterogeneous soft partitions and their unequal trustworthiness. Rough sets and granular computing [128] then introduced upper and lower approximations, allowing uncertainty to be aggregated across multiple granularity levels [127], [130]. Three-way ensemble clustering [135], [177] made deferred decision-making central by identifying boundary samples whose assignments should remain cautious rather than forced [178]. Dempster–Shafer evidence theory [123] provides another fusion language, in which ensemble members contribute belief rather than only votes or memberships [129].

The motivation is well established, but calibration and validation remain less developed. Fuzzy memberships, rough approximations, and evidence masses are useful only if they provide interpretable confidence, change the final decision process, or improve robustness under uncertainty. Future progress will depend on making uncertainty an operational part of consensus formation, rather than a descriptive quantity attached after a hard clustering decision has already been made.

C. Representation and Scalability Methods

Classical ensemble clustering faces two practical challenges beyond consensus construction and reliability estimation: the scalability bottleneck of dense pairwise methods on large datasets, and the limited expressiveness of hand-crafted ensemble features on high-dimensional or complex data. This category therefore addresses two complementary questions: how to learn a compressed consensus representation that scales efficiently, and how to construct an enhanced representation that improves ensemble generation or consensus learning.

1) *Representation Enhancement Methods*: Representation enhancement methods focus on the quality of the evidence supplied to consensus learning. Classical ensemble clustering often fuses base partitions, but these representations may be too limited for high-dimensional or highly nonlinear data. The goal is to construct an intermediate latent consensus-aware space in which ensemble generation or aggregation becomes easier and more faithful.

This issue is increasingly important on modern data, because even a strong consensus rule cannot recover structure that is absent from the evidence it receives. One line of work improves the feature space before fusion. Autoencoder-based models [179], [155], [23] and multi-view latent fusion [180] all follow this logic by making the generated base partitions more informative and robust before the ensemble stage begins

[158]. In such methods, representation quality is treated as a prerequisite for reliable consensus rather than as a separate preprocessing detail.

Later work moves beyond feature generation and treats consensus learning itself as a representation-design problem. Instead of handing a latent feature space to a fixed consensus function, these methods explicitly learn sample-level, cluster-level, or partition-level representations that are better aligned with the target consensus. Attention, graph learning, bipartite modeling, and confidence-guided aggregation are representative mechanisms here, because they try to encode which relations in the ensemble should be emphasized and which should be suppressed. ECAR [157] is a representative example in this direction, using an attentional encoder to model sample importance with respect to group-level structures induced by the ensemble.

A further step is end-to-end joint optimization, where representation learning, cluster alignment, and consensus fusion are optimized within one differentiable or tightly coupled framework. Such methods aim to let consensus quality directly refine the learned representation rather than treating the two stages separately [181], [159]. There is also an inverse trend in which ensemble structures themselves serve as supervision for downstream representation learning tasks, for example in subspace learning [156] or cross-modal hashing [182]. The category therefore broadens the classical consensus question: not only how should partitions be fused, but also in what space should fusion take place?

Representation enhancement methods can improve the evidence used for consensus, but they also add extra modeling choices and reduce interpretability. A central risk is that the representation module may dominate the pipeline, making the method closer to representation learning or deep clustering than to ensemble clustering. Future work in this direction is most convincing when it can clearly show why ensemble information is necessary, and how it improves the learned representation rather than merely increasing model complexity.

2) *Scalable Ensemble Methods*: Scalable ensemble methods target the computational bottleneck of classical consensus learning. Many strong CA-matrix and graph-based approaches require $O(n^2)$ pairwise modeling and expensive spectral decomposition, which makes them difficult to use on large datasets. Scalable methods therefore replace full pairwise consensus with compressed structures such as representatives, anchors, micro-clusters, or sparse graphs, while trying to retain enough global information for accurate aggregation.

The practical role of this family is to make ensemble clustering usable beyond small and medium-sized datasets. Representative sampling [150], [154] and compressed consensus structures [142] reduce the number of entities that must be fused directly [149], [146]. Landmark-based [143], anchor-based [153], and approximation-based methods [147] reduce the cost of graph learning and spectral partitioning by replacing dense constructions with sparse surrogate structures or one-shot decompositions [183], [144]. When the bottleneck is system-level rather than algorithmic, distributed [184], privacy-preserving [185], and hardware-oriented frameworks [186] further extend scalability by localizing ensemble con-

struction and limiting global communication [187].

The price of scalability is approximation. Once consensus is formed over representatives or sparse surrogates rather than over all sample pairs, rare structures, minority clusters, or weak long-range relations may be lost. These methods also introduce a new layer of design decisions, such as how to choose anchors, how many representatives to keep, or how to distribute local summaries. Scalable methods therefore do not eliminate the difficulty of consensus learning; they move part of it into the design of the compressed representation.

The line has progressed from prototype extraction and subset-based aggregation toward more principled approximation. Anchor graphs, sparse bipartite formulations, and one-pass or near-linear spectral surrogates make the compressed representation less ad hoc, while distributed optimization, privacy preservation, and hardware acceleration extend the question to deployment settings. For this reason, scalable methods are better viewed as a systems and approximation layer over ensemble clustering than as a single consensus philosophy. Their theory remains underdeveloped: sampling, anchor, representative, or distributed variants are most convincing when they clarify what ensemble-specific information is preserved and what is lost. Future work should emphasize approximation guarantees, adaptive representative learning based on reliability or uncertainty, streaming updates, federated and privacy-aware consensus, and large-scale deployment settings with explicit memory and communication constraints.

VI. APPLICATIONS OF ENSEMBLE CLUSTERING

Ensemble clustering has been used in many domains where data naturally support multiple plausible partitions. Its value is not limited to averaging unstable runs. More broadly, it provides a way to combine evidence from different resolutions, modalities, representations, or perturbations. Based on the collected literature, its applications can be grouped into five representative domains, as shown in Fig. 4. These domains cover omics and molecular biomedical analysis, healthcare analytics and clinical decision support, temporal and spatiotemporal monitoring, vision, remote sensing, and multimedia analysis, as well as textual, network, and behavioral analytics. Together, they show that ensemble clustering is particularly useful for high-dimensional, heterogeneous, noisy, dynamic, or multi-representation data.

A. Omics and Molecular Biomedical Analysis

Omics-driven biomedical analysis, including gene expression profiling, DNA microarray analysis, tumor molecular profiling, and single-cell transcriptomics, has been one of the most active application areas for ensemble clustering. Biological structure is often multi-resolution. One clustering may reveal broad lineages, while another captures fine-grained cell states or tumor subtypes. Ensemble clustering is useful because it integrates such complementary partitions without forcing the analysis to rely on a single resolution.

Early studies focused on class discovery and subgroup identification in gene expression data. Graph-based consensus clustering improved class discovery from gene expression

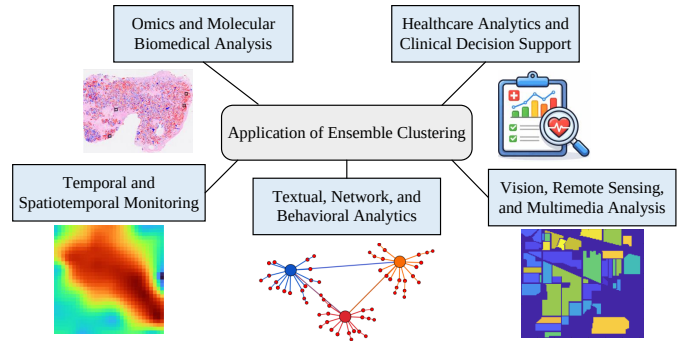


Fig. 4. Important application domains of ensemble clustering.

profiles [7], while link-based and multi-objective ensemble methods enhanced the stability of gene expression clustering [54], [100]. Individual clustering and homogeneous ensemble strategies also showed the benefit of combining multiple partitions in noisy biological data [188]. This line was extended to microarray-based molecular analysis, where fuzzy ensemble clustering and random projection based strategies were used for high-dimensional DNA microarray data [189], [190]. Predictive weighted ensembles assigned different importance to base results in microarray clustering [191]. Tumor-related molecular profiling also adopted ensemble strategies, such as semi-supervised clustering ensembles for tumor clustering from gene expression profiles [192].

Recent work has shifted toward single-cell and spatial multi-omics data, where sparsity, technical noise, and biological heterogeneity make clustering sensitive to preprocessing and resolution choices. Aggregated ensemble clustering and evolutionary multi-objective ensemble pruning have been used for single-cell RNA-seq analysis [40], [109]. Probabilistic graphical models and Bayesian-network based ensemble methods further model relationships among candidate partitions [193], [194]. Low-rank subspace ensemble learning, multi-objective particle optimization, and ensemble feature selection have also improved cell-type detection [195], [196], [197]. Recent studies extend this direction through deep denoising autoencoders, cross-tabulation based consensus, high-order element-wise weighting, and spatial multi-omics ensemble learning [198], [199], [200], [201]. Overall, this domain remains prominent because ensemble clustering matches the way biological evidence is often interpreted, as partially overlapping signals rather than one fixed partition.

B. Healthcare Analytics and Clinical Decision Support

Healthcare-oriented analysis, including medical diagnosis, patient stratification, medical image analysis, and biomedical time-series modeling, forms another major application domain. Clinical data are often heterogeneous, small in subtype-specific cohorts, and affected by missing or noisy measurements. In this context, the key requirement is not only accuracy but also replicability. Patient groups should not change drastically because of initialization, feature selection, or preprocessing. Ensemble clustering supports this goal by pooling evidence across multiple clustering runs.

A substantial body of work addresses medical diagnosis and patient stratification. Early decision-support pipelines combined clustering ensembles with neural network ensembles for knowledge discovery [202]. For patient stratification, evolutionary multi-objective clustering and link-weight based ensemble methods were used to obtain more stable patient groups [8], [203]. Clinical information has also been incorporated to infer robust tumor subtypes [204], while integrative hierarchical ensemble clustering has supported disease subtype discovery [205].

Cancer-oriented studies form a notable subfield. Adaptive fuzzy consensus clustering and extended fuzzy k -means ensembles were developed for cancer data analysis [126], [134]. Other studies analyzed ensemble algorithms for cancer patient data and cancer clustering tasks [206], [207]. More recent work has explored double weighted ensemble clustering and granular-computing based strategies for cancer subtype analysis and cervical cancer predictor extraction [208], [209]. Medical image analysis is another active subcategory, with applications in carotid artery segmentation, MRI brain image segmentation, and histopathological image analysis [124], [210], [211]. Fuzzy ensemble methods based on KL-divergence also demonstrate the use of consensus learning in medical image segmentation [212]. Biomedical time-series modeling, although less represented, shows the value of ensemble methods in clinical monitoring with noisy physiological signals [213]. Across these tasks, ensemble clustering serves as a stabilizing layer between noisy measurements and clinical subgroup analysis.

C. Temporal and Spatiotemporal Monitoring

Temporal and spatiotemporal monitoring covers time-series modeling, industrial condition monitoring, environmental assessment, energy system analysis, and mobility pattern discovery. For dynamic systems, representation is itself a modeling choice. The same signal may be described in the time domain, frequency domain, symbolic sequences, or state-space form. Ensemble clustering can fuse partitions from these complementary representations and reduce the risk of missing important dynamics under one encoding.

Many studies in this domain focus on time-series and sequential data. HMM-based hybrid meta-clustering ensembles and adaptive bi-weighting variants improved temporal clustering under model-selection and initialization uncertainty [9], [111], [38]. Weighted clustering ensembles with different temporal representations, competitive learning networks, and RPCL network ensembles further showed how multiple sequence descriptions can support robust temporal clustering [214], [215], [216]. Recent studies extend this line to multivariate time-series clustering and soft subspace based ensemble clustering [70], [217].

Industrial monitoring and fault diagnosis form another important subgroup. Ensemble clustering has been used to combine traditional and deep representation features for fault diagnosis [180], identify operational modes in industrial gas turbines [218], and improve photovoltaic fault diagnosis [219]. Similar ideas have been applied to railway signaling fault

risk assessment and power plant performance assessment [220], [221]. Other applications include air quality monitoring network assessment [222], explainable clustering for energy systems [223], and traffic trajectory or mobility analysis [224], [158]. In these tasks, the consensus partition helps reconcile several partial descriptions of a dynamic process.

D. Vision, Remote Sensing, and Multimedia Analysis

Visual and image-centered problems form another important application area, covering image segmentation, computer vision, and remote sensing image analysis. Image segmentation is often underdetermined. Multiple boundary configurations may be plausible, and small changes in initialization, parameters, or features can alter the resulting regions. Ensemble clustering aggregates these candidate segmentations to produce boundaries supported by repeated evidence.

In image segmentation and computer vision, ensemble strategies have been applied to unsupervised image partitioning, region discovery, visual categorization, and image security analysis. Early work used ensemble clustering for visual region detection [225], [226]. Other studies applied ensemble strategies to automatic image segmentation, unsupervised visual categorization, and image partitioning [227], [228], [229]. Infinite ensemble clustering and unsupervised image segmentation further show how multiple candidate partitions can reduce instability [10], [230]. Recent applications include JPEG steganalysis, geometric ensemble clustering, and joint segmentation or representation learning for visual data [231], [232], [181].

Remote sensing image analysis applies ensemble clustering to SAR and hyperspectral imagery, where high dimensionality, spectral mixing, and spatial heterogeneity make land-cover boundaries hard to infer from a single partition. Spectral ensemble clustering and multi-objective ensemble strategies have been used for remote sensing image segmentation and classification [233], [234]. Later studies applied consensus construction, improved clustering frameworks, and correlation-based ensemble methods to remote sensing images [235], [236], [237]. In both natural and remote sensing imagery, ensemble clustering allows spatial, spectral, and algorithmic evidence to reinforce one another.

E. Textual, Network, and Behavioral Analytics

Ensemble clustering has also been applied to text, user behavior, network structure, and security-related pattern discovery. These tasks share a common feature: the same object can be represented through several digital traces. Documents may be encoded as bags of words, topics, or neural embeddings; user behavior may appear as transaction sequences, interaction graphs, or profiles; and networks may be described by adjacency matrices, spectral embeddings, or diffusion patterns. Ensemble clustering integrates these partial views into consensus groupings that are less tied to any single representation.

Text mining and document clustering constitute the largest subgroup. Early studies used feature subset based and ensemble-based document clustering to improve robustness

in text grouping [238], [239]. Multi-view clustering and sentiment-oriented ensemble methods were later applied to documents with heterogeneous textual representations [240], [241]. Other applications include semantic document grouping, ontology alignment, comparative document clustering, and bibliometric-text mining integration [242], [11], [243], [244].

Cybersecurity and anomaly detection use ensemble clustering for malware categorization, anomaly discovery, and attack detection, where consensus can reduce dependence on fragile representations. Ensemble clustering has been applied to anomaly detection and malware family analysis [114], [55]. Deep and graph-related ensemble strategies further support attack detection and security-oriented pattern discovery [22], [245], [246]. Recommender systems and user profiling also benefit from ensemble strategies that combine behavioral, relational, and content-based evidence [247], [59], [248], [249]. Social network analysis and business process management represent additional subcategories, where ensemble clustering helps stabilize community discovery and process-oriented grouping [250], [107], [251]. Across these tasks, ensemble clustering provides a principled way to integrate complementary representations that a single clustering method cannot fully exploit.

VII. EXPERIMENTS

In this section, we provide a comprehensive experimental benchmark of representative ensemble clustering methods. The goal is not to propose a new algorithm, but to offer a systematic and fair comparison across different methodological categories, thereby revealing their empirical characteristics, strengths, and limitations.

A. Experimental Settings

All experiments were implemented in MATLAB R2024a and executed on a workstation with an Intel(R) Xeon(R) Gold 6248R CPU @ 3.00GHz and 767 GB RAM. To ensure reproducibility, the random seed was fixed using `rng(2026)`. Each experiment was independently repeated ten times, and the results are reported as the average performance. Hyperparameters were tuned within the ranges suggested in the original paper, as detailed in the supplementary material (however, for some computationally expensive methods, we narrowed the search space accordingly).

B. Experimental Data

We collected a total of 86 datasets spanning diverse domains, including image analysis, text mining, and biological data. These datasets cover a wide range of data characteristics, such as varying sample sizes, feature dimensions, cluster structures, and noise levels, providing a comprehensive testbed for evaluating the performance and robustness of ensemble clustering methods. Detailed information about these datasets is provided in the supplementary material.

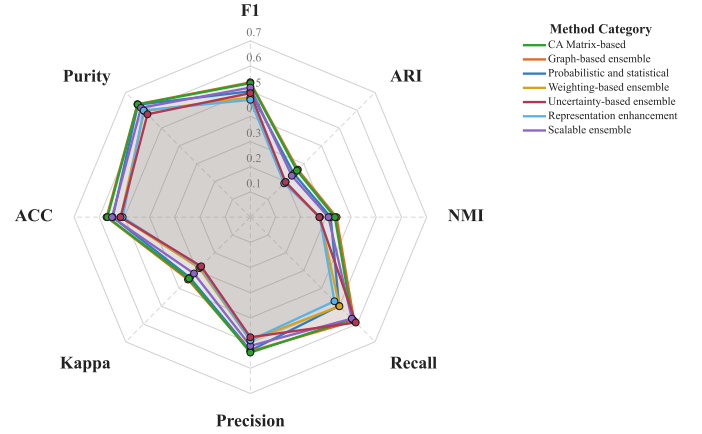


Fig. 5. Comparison of different ensemble clustering method categories across multiple metrics

C. Evaluation Methods

The benchmark includes 76 method variants selected from all seven subcategories of our taxonomy. For each subcategory, we selected representatives that are widely cited, span different publication periods, and reflect the diversity of technical approaches within that subcategory. The methods were published in venues such as ICDM, ICML, IJCAI, AAAI, IEEE TPAMI, IEEE TKDE, IEEE TNNLS, Information Fusion, and Pattern Recognition. All methods were evaluated under a unified protocol: the same base clustering generation procedure, the same 86 datasets, and the same eight external metrics.

D. Evaluation Metrics

To comprehensively evaluate the performance of different ensemble clustering methods, we adopt eight widely used external evaluation metrics, namely Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), F1-score (F1), Purity, Accuracy (ACC), Cohen’s Kappa (Kappa), Precision, and Recall. These metrics assess clustering quality from different perspectives, including information consistency, pairwise agreement, classification accuracy, and class-level retrieval performance.

E. Experiment Analysis

To provide a comprehensive empirical evaluation, we collected representative methods from all major ensemble clustering categories and compared them under a unified experimental protocol. Table III reports the average results over all 86 datasets for the 76 benchmarked method variants, while Fig. 5 provides a subcategory-level radar comparison of the same benchmark.

The results show that graph-based and CA matrix-based methods achieve strong and stable performance across most metrics, confirming the effectiveness of directly modeling consensus structures through graph relations or pairwise co-association. Probabilistic and scalable ensemble methods also obtain competitive results, suggesting that statistical aggregation and compressed consensus representations can preserve useful ensemble information. In contrast, weighting-based,

TABLE III
AVERAGE PERFORMANCE OF EACH METHOD ON EIGHT METRICS.

Category	Method	Venue	NMI	ARI	F1	Purity	ACC	Kappa	Precision	Recall
CA Matrix-based methods	[66]	Information Fusion	0.3648	0.2924	0.5473	0.6430	0.5824	0.3735	0.5463	0.5884
	[162]	AAAI	0.3886	0.3241	0.5761	0.6695	0.6304	0.4202	0.5588	0.6166
	[162]	AAAI	0.3610	0.2828	0.5317	0.6536	0.5848	0.3727	0.5405	0.5459
	[68]	AAAI	0.1158	0.0586	0.3469	0.4990	0.3853	0.1370	0.3962	0.3232
	[68]	AAAI	0.4045	0.3434	0.6107	0.6703	0.6231	0.3988	0.5819	0.7171
	[6]	IEEE TPAMI	0.3531	0.2794	0.5446	0.6404	0.5791	0.3617	0.5398	0.5846
	[73]	IEEE TNNLS	0.3782	0.3227	0.5832	0.6609	0.6188	0.4041	0.5613	0.6435
	[252]	Neurocomputing	0.3631	0.2895	0.5432	0.6511	0.5818	0.3718	0.5550	0.5669
	[72]	ACM MM	0.4570	0.4086	0.6180	0.7324	0.6504	0.4754	0.6698	0.5913
	[69]	AAAI	0.3194	0.2389	0.5224	0.6122	0.5519	0.3169	0.5030	0.6040
	[54]	Bioinformatics	0.3714	0.2976	0.5423	0.6564	0.5831	0.3834	0.5622	0.5518
	[161]	ACM CIKM	0.3333	0.2634	0.5158	0.6454	0.5637	0.3574	0.5486	0.5084
	[161]	ACM CIKM	0.3182	0.2361	0.5023	0.6349	0.5507	0.3320	0.5226	0.5059
	[71]	IEEE TKDE	0.3539	0.2924	0.5690	0.6420	0.6018	0.3759	0.5366	0.6431
	[62]	IEEE TKDE	0.2713	0.1598	0.4947	0.5863	0.5215	0.2432	0.4521	0.5946
	[75]	IEEE TCSVT	0.3364	0.2694	0.5003	0.6456	0.5658	0.3888	0.5463	0.4719
	[60]	Pattern Recognition	0.2943	0.1841	0.5033	0.5981	0.5284	0.2711	0.4661	0.6148
	[253]	LNCS	0.3237	0.2444	0.4993	0.6412	0.5461	0.3381	0.5393	0.4847
	[254]	IJCAI	0.3397	0.2582	0.5124	0.6374	0.5561	0.3470	0.5257	0.5257
	[255]	ACM TKDD	0.3249	0.2417	0.5342	0.6094	0.5604	0.3193	0.6162	0.4979
	[255]	ACM TKDD	0.3422	0.2627	0.5383	0.6264	0.5746	0.3483	0.5967	0.5155
	[240]	JIIS	0.1747	0.0938	0.4997	0.4975	0.4743	0.1338	0.4002	0.8484
Graph-based ensemble methods	[46]	IEEE TKDE	0.3202	0.2370	0.4934	0.6292	0.5426	0.3322	0.5276	0.4854
	[43]	Information Fusion	0.3313	0.2661	0.5452	0.6172	0.5713	0.3462	0.5156	0.6412
	[256]	Neurocomputing	0.3598	0.2868	0.5500	0.6462	0.5858	0.3696	0.5455	0.5904
	[256]	Neurocomputing	0.3693	0.2950	0.5422	0.6528	0.5812	0.3780	0.5582	0.5583
	[42]	IEEE TSMCS	0.3669	0.2950	0.5597	0.6368	0.5869	0.3691	0.5349	0.6328
	[42]	IEEE TSMCS	0.3685	0.2956	0.5489	0.6495	0.5867	0.3795	0.5537	0.5821
	[36]	IEEE TKDE	0.3388	0.2559	0.5277	0.6369	0.5632	0.3387	0.5296	0.5619
	[36]	IEEE TKDE	0.3259	0.2393	0.5140	0.6292	0.5504	0.3249	0.5252	0.5376
	[36]	IEEE TKDE	0.2277	0.1323	0.5070	0.5418	0.5055	0.1934	0.4278	0.7424
	[36]	IEEE TKDE	0.3515	0.2739	0.5241	0.6414	0.5645	0.3591	0.5434	0.5399
	[163]	ACM TKDD	0.3579	0.2923	0.5601	0.6416	0.5967	0.3777	0.5340	0.6227
	[28]	ICML	0.3662	0.2935	0.5412	0.6539	0.5835	0.3801	0.5606	0.5508
	[44]	AAAI	0.3596	0.2961	0.5606	0.6452	0.5991	0.3804	0.5418	0.6053
	[47]	IJCAI	0.3493	0.2765	0.5205	0.6573	0.5752	0.3752	0.5579	0.5085
Probabilistic and statistical methods	[78]	ICDM	0.3173	0.2439	0.4869	0.6318	0.5373	0.3444	0.5338	0.4693
	[18]	IEEE TPAMI	0.3291	0.2558	0.4953	0.6382	0.5481	0.3589	0.5429	0.4768
	[81]	IEEE TPAMI	0.3554	0.2811	0.5224	0.6519	0.5684	0.3765	0.5577	0.5212
	[166]	Pattern Recognition	0.3450	0.2691	0.5362	0.6404	0.5734	0.3548	0.5381	0.5666
	[94]	IEEE TPAMI	0.2420	0.1751	0.4404	0.6018	0.4901	0.2745	0.5007	0.4090
	[257]	ICML	0.3384	0.2731	0.5234	0.6437	0.5788	0.3731	0.5389	0.5217
	[90]	IEEE TKDE	0.3051	0.2101	0.4939	0.6119	0.5286	0.2934	0.4947	0.5231
Weighting-based ensemble methods	[102]	Pattern Recognition Letters	0.2275	0.1532	0.4188	0.5526	0.4584	0.2418	0.4557	0.4187
	[20]	IEEE TNNLS	0.2230	0.1366	0.4409	0.5599	0.4888	0.2343	0.4362	0.4826
	[113]	IEEE TCYB	0.3777	0.3151	0.5676	0.6605	0.6049	0.3986	0.5637	0.6088
	[113]	IEEE TCYB	0.3822	0.3173	0.5615	0.6684	0.6040	0.4025	0.5715	0.5809
	[117]	IEEE TCYB	0.2919	0.1887	0.5007	0.6019	0.5300	0.2699	0.4751	0.5846
	[117]	IEEE TCYB	0.2993	0.2070	0.4926	0.6151	0.5328	0.2932	0.4982	0.5132
	[117]	IEEE TCYB	0.3136	0.2183	0.5031	0.6227	0.5423	0.3032	0.5043	0.5308
	[98]	Information Fusion	0.3454	0.2672	0.5386	0.6365	0.5722	0.3527	0.5358	0.5789
	[258]	IEEE TNNLS	0.3012	0.2289	0.5149	0.5914	0.5394	0.3083	0.4837	0.5977
	[101]	Pattern Recognition	0.1218	0.0774	0.3700	0.5040	0.4014	0.1507	0.4048	0.3788
	[259]	ICDM	0.3277	0.2486	0.4917	0.6372	0.5432	0.3545	0.5415	0.4777
	[259]	ICDM	0.2232	0.1634	0.4287	0.5677	0.4730	0.2575	0.4783	0.4237
	[259]	ICDM	0.2970	0.2225	0.4678	0.6257	0.5240	0.3305	0.5305	0.4352
	[260]	Bioinformatics	0.3397	0.2548	0.5370	0.6381	0.5698	0.3369	0.5275	0.5820
Uncertainty-based ensemble methods	[261]	Applied Artificial Intelligence	0.3209	0.2407	0.4910	0.6313	0.5390	0.3377	0.5252	0.4838
	[189]	Artificial Intelligence in Medicine	0.3095	0.2255	0.4910	0.6185	0.5378	0.3194	0.5129	0.4878
	[127]	KBS	0.2344	0.1570	0.4952	0.5287	0.4917	0.2140	0.4304	0.7011
	[129]	Information Sciences	0.2874	0.2182	0.4916	0.5916	0.5228	0.3007	0.4903	0.5504
	[176]	FSS	0.3545	0.2809	0.5458	0.6418	0.5807	0.3634	0.5415	0.5855
	[137]	IEEE TFS	0.2042	0.1304	0.4668	0.5165	0.4677	0.1975	0.4187	0.6697
	[137]	IEEE TFS	0.2060	0.1295	0.4652	0.5162	0.4681	0.1994	0.4182	0.6636
Representation enhancement methods	[159]	IEEE TNNLS	0.3395	0.2654	0.5181	0.6334	0.5578	0.3503	0.5372	0.5294
	[262]	PR	0.2819	0.1901	0.4670	0.6069	0.5074	0.2837	0.4945	0.4684
	[198]	Bioinformatics	0.3566	0.2809	0.5315	0.6424	0.5697	0.3636	0.5479	0.5474
	[181]	ICME	0.2348	0.1534	0.4278	0.5789	0.4758	0.2549	0.4686	0.4145
	[263]	Information Sciences	0.2584	0.1649	0.4541	0.5944	0.4936	0.2580	0.4719	0.4672
	[158]	IEEE TITS	0.1833	0.0848	0.3962	0.5384	0.4377	0.1701	0.4135	0.4058
Scalable ensemble methods	[142]	Pattern Recognition	0.3038	0.2295	0.4908	0.6148	0.5330	0.3205	0.5148	0.5015
	[146]	IJAR	0.2891	0.2035	0.4960	0.5902	0.5319	0.2921	0.4770	0.5837
	[264]	Artificial Intelligence Review	0.1915	0.0965	0.4834	0.5249	0.4812	0.1581	0.4077	0.6874
	[153]	Information Fusion	0.3469	0.2798	0.5402	0.6497	0.5877	0.3684	0.5458	0.5534
	[151]	IEEE TKDE	0.3643	0.2919	0.5369	0.6520	0.5801	0.3792	0.5595	0.5442
	[148]	IEEE TKDE	0.3655	0.2940	0.5396	0.6537	0.5818	0.3806	0.5614	0.5478

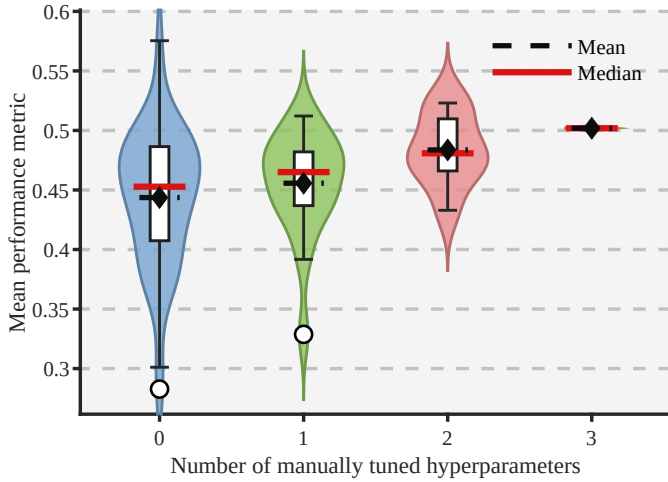


Fig. 6. Relationship between tunable hyperparameter count and benchmark performance across the 76 method variants. ‘None’ and fixed values are counted as zero, whereas each searched range or candidate set is counted as one tunable hyperparameter.

uncertainty-based, and representation enhancement methods show more diverse performance, reflecting their stronger dependence on reliability estimation, uncertainty modeling, or representation quality.

Fig. 5 further summarizes the benchmark results from a category-level perspective. The radar chart shows that graph-based and CA matrix-based methods form the outer envelope on most evaluation axes, indicating their overall advantage among the seven subcategories. Probabilistic and scalable methods constitute a relatively stable middle group, while ensemble weighting and representation enhancement methods are less dominant on average. Uncertainty-based methods exhibit a distinctive profile, especially on recall, suggesting that explicit uncertainty modeling may help preserve ambiguous or difficult assignments.

Fig. 6 further examines the relationship between the number of manually tuned hyperparameters and benchmark performance. The results show a generally positive but moderate trend: methods with more tunable hyperparameters tend to achieve slightly higher mean and median performance. Specifically, the average performance increases from methods with no tuned hyperparameters to those with one, two, and three tuned parameters. However, the improvement is not dramatic, suggesting that larger hyperparameter search spaces alone cannot fully explain the performance differences among methods. Instead, the underlying consensus mechanism, reliability modeling strategy, and representation design remain the primary factors affecting clustering quality. This observation also indicates that overly complex parameter tuning is not necessarily a guarantee of better performance, and practical ensemble clustering methods should balance accuracy, robustness, and ease of use.

We also record the running time of all evaluated methods on all benchmark datasets to provide a more complete view of their computational behavior. Due to page limitations, the full runtime matrix and detailed runtime statistics are reported in the supplementary material.

VIII. FUTURE DIRECTIONS

Although ensemble clustering has achieved substantial progress in many aspects, several important challenges remain open. In this section, we highlight several promising directions that may advance the development of ensemble clustering.

A. How to Determine the Number of Clusters k

Determining the number of clusters k remains a fundamental challenge in ensemble clustering, because the consensus process must integrate multiple base partitions that may be produced by different algorithms, parameter settings, and structural assumptions. In many existing methods, k is specified in advance or fixed across all base clusterings, which simplifies consensus learning but limits the flexibility of the framework. Different base clusterings may reveal complementary structures at different levels of granularity, and enforcing a single predefined k may obscure such information. A promising direction is to estimate k adaptively from the ensemble itself, rather than treating it as an external input. More flexible consensus formulations that support hierarchical or multi-scale cluster structures would also help the final partition better reflect the heterogeneous information contained in the ensemble.

B. How to Enhance Diversity in the Ensemble

Diversity is a key factor in the success of ensemble clustering, since the quality of the consensus depends on whether the base clusterings provide complementary rather than redundant information. Existing methods often introduce diversity through random initialization, parameter perturbation, resampling, feature subspace selection, or heterogeneous clustering algorithms. However, such diversity is usually generated heuristically and is rarely evaluated in relation to the final consensus quality. As a result, an ensemble may contain partitions that are either overly similar or highly variable but unreliable, both of which reduce the benefit of aggregation. Future research should move beyond simply increasing randomness and instead focus on constructing effective diversity for ensemble clustering. This includes developing principled diversity measures at the partition, cluster, and sample levels, jointly considering diversity and quality during ensemble generation, and designing adaptive selection or weighting strategies that preserve informative complementarity while suppressing noisy or low-value members.

C. Stability and Robustness of Ensemble Clustering

Despite its motivation of improving reliability over single clustering methods, ensemble clustering can still be sensitive to small perturbations in the data, the ensemble composition, or the generation of base partitions. Variations caused by sampling, noise, initialization, feature perturbation, or low-quality clusterings may propagate through the consensus process and lead to noticeably different results. This issue is especially important in ensemble clustering because the final partition is inferred solely from multiple clustering outputs, making it difficult to distinguish stable structural patterns

from accidental agreement. Stability and robustness should therefore be treated as properties of the entire ensemble pipeline, including base clustering generation, weighting or selection, and consensus construction. Promising directions include developing perturbation-insensitive consensus functions, incorporating uncertainty-aware or reliability-aware aggregation schemes, using resampling-based consistency analysis, and explicitly identifying ambiguous or boundary samples whose assignments remain unstable across the ensemble.

D. Interpretability of Ensemble Clustering

Interpretability is becoming increasingly important in ensemble clustering, especially when consensus results are used in scientific analysis or decision-critical scenarios. Although ensemble clustering can improve performance by aggregating multiple base partitions, the final consensus is often produced through complex interactions among co-association structures, graph relations, weighting mechanisms, or probabilistic models. As a result, it is often unclear why certain samples are grouped together, why one consensus solution is preferred over others, or how different base clusterings contribute to the final partition. Future research should therefore aim to improve interpretability at multiple levels, including the roles of individual base partitions, clusters, and samples in shaping the consensus result. More transparent ensemble clustering frameworks would not only enhance user trust and practical usability, but also provide deeper insight into the behavior and limitations of existing methods.

E. Fairness in Ensemble Clustering

Fairness is an emerging but still underexplored issue in ensemble clustering, particularly when clustering results are applied to socially sensitive or decision-critical tasks. Although the ensemble process is intended to improve robustness through aggregation, it may also inherit or even amplify biases already present in the base clusterings, especially when the data exhibit imbalanced group structures, skewed feature distributions, or latent sensitive attributes. In such cases, the consensus partition may systematically favor dominant patterns while neglecting minority structures, resulting in distorted or unfair outcomes. Fairness needs to be considered throughout the ensemble clustering pipeline, including base clustering generation, ensemble selection, weighting, and consensus construction. Important directions include designing fairness-aware consensus objectives, balancing the influence of majority and minority groups during aggregation, and studying the trade-offs among fairness, diversity, and clustering quality.

IX. CONCLUSION

In this paper, we presented a comprehensive survey of ensemble clustering from the perspectives of problem formulation, methodological taxonomy, application domains, experimental benchmarking, and future challenges. We organized existing studies into three major categories, namely Consensus Modeling Methods, Ensemble Reliability Methods, and Representation and Scalability Methods, and discussed

several important problem settings that have received limited attention in previous surveys, including the relation to multi-view clustering, the use of original features, and incomplete-data scenarios. In addition, we conducted a large-scale empirical study by comparing 76 method variants on 86 datasets across eight evaluation metrics. The results highlight both the strong performance of classical paradigms and the potential of emerging directions such as scalable ensemble methods, representation enhancement, and uncertainty-aware ensemble clustering. We hope this survey can serve as a useful reference for future research and the development of next-generation ensemble clustering methods.

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